Independent projectile directions

antocciphysics | TEACHER KEY | CPA

Learning target: Explain why horizontal and vertical projectile motion share time but evolve differently.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 6; 3 course days later.*
g=5 N/kg; force 35 N toward the source.

02. Explain a worked example (worked)
Horizontal arrow stays 3 m/s; vertical arrow begins zero and grows to 20 m/s downward.

03. Look and explain (model)
Same fall time if same initial vertical velocity and negligible air resistance.

04. Build the solution (practice)
6 m; range doubles to 12 m, fall time unchanged.

05. Use the idea (reason)
Horizontal velocity alone does not change vertical acceleration; lift/another vertical interaction would be required, outside this ideal model.

06. Choose a method (mixed) *Review: Unit 1, Day 19; 12 course days later.*
The student multiplied by mass. Divide: $a=F_{net}/m=6$ m/s². Check units N/kg=m/s² or substitute ma back into F_{net}.

07. Exit check (exit)
Equal times; second range 3 times first.

08. STRETCH: Apply it another way (stretch)
Same fall time; ranges 6 and 15 m. Faster horizontal motion does not delay falling in the ideal model.

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01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 6; 3 course days later.*
Field 3 N/kg; force 2.1 N toward the source.

02. Explain a worked example (worked)
Horizontal arrow stays 3 m/s; vertical arrow begins zero and grows to 20 m/s downward.

03. Represent and interpret (model)
Same fall time if same initial vertical velocity and negligible air resistance.

04. Solve and justify (practice)
Δx 12 m;Δy=-11.25 m; v=(8,-15) m/s.

05. Analyze the reasoning (reason)
Horizontal velocity alone does not change vertical acceleration; lift/another vertical interaction would be required, outside this ideal model.

06. Choose a method (mixed) *Review: Unit 1, Day 19; 12 course days later.*
Central 52 m; half-width 5 m; interval [47, 57] m. Valid checks include units and upper/lower endpoint substitution.

07. Exit check (exit)
Speed 10 m/s; need initial position and elapsed motion/history.

08. STRETCH: Apply it another way (stretch)
(6,-2) m/s; horizontal remains constant without horizontal force. Both motions share elapsed time.